

“Less Is More”: Decision-Making under Increasingly Complex Ambiguity

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Abstract

This paper examines whether increasingly complex decisions under ambiguity are better described by more complex behavioural models. Using an allocation experiment, we vary complexity through the number of possible outcomes and decision stages and compare Expected Utility (EU), α MaxMin (AMM), and Choquet Expected Utility (CEU). EU is the most frequently selected model according to AICc in all three ambiguous-lottery parts, although its relative advantage varies across tasks and CEU remains competitive in the most complex part. Estimated risk-attitude parameters are relatively stable across tasks, while ambiguity parameters vary more and become more dispersed in the more complex parts. These results suggest that additional ambiguity parameters do not uniformly improve penalized model fit as task structure becomes more complex. Cognitive difficulty and decision noise are possible explanations, but the experimental design does not distinguish these mechanisms from heterogeneity in ambiguity attitudes or differences in parameter identification. The findings therefore highlight the importance of evaluating model complexity empirically rather than assuming that structurally more complex decisions necessarily require more parameterized behavioural models.

JEL codes: D81, D91, C91

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1. Introduction

Consider an investor allocating a retirement portfolio in a volatile market. They must weigh familiar risks, where outcomes have known probabilities, such as the historical volatility of a stock index, against deeper ambiguities, where the probabilities are fundamentally unknown. Is a new geopolitical conflict 10% likely or 50%? Will a breakthrough in technology succeed in three years or ten? Facing this complex mix of risky and ambiguous prospects, does the investor employ a sophisticated mental calculus that takes this “unknown-ness” into account? Or, overwhelmed by complexity, does the investor simplify, perhaps treating ambiguous prospects as if they were merely risky? Such a scenario highlights a critical distinction in decision theory between risk and ambiguity. It also frames the central research question of this paper: As choice problems become more complex, do we need more complex models to describe behaviour under ambiguity, or can relatively simple models provide an equally good or better empirical description? This question is particularly relevant because real-world decisions are often ambiguous and multi-stage, whereas behavioural models are commonly evaluated in simpler settings.

To investigate this question, we conduct a laboratory experiment using an allocation method. We examine how individuals allocate endowments across ambiguous lotteries of increasing structural complexity, in which they are informed of two possible values of the objective probability of each ambiguous event: a lower probability and a higher probability. Complexity varies along two dimensions: the number of possible outcomes and the number of decision stages. We compare three models: Expected Utility (EU), which does not explicitly incorporate ambiguity attitudes, and the more flexible α MaxMin (AMM) and Choquet Expected Utility (CEU) models. Comparing their goodness-of-fit allows us to examine how the relative empirical performance of these models changes across differently structured ambiguous choice problems.

Counter to the intuition that increasingly complex ambiguous problems necessarily require increasingly complex behavioural models, our results provide qualified support for a “less is more” pattern. EU is the most frequently selected model according to AICc in all three ambiguous-lottery parts, although the strength of its relative advantage differs across task structures. The evidence is strongest in Part ALB, while CEU remains competitive with EU in Part ALC. The estimated risk-aversion parameter under EU is also relatively stable across parts, whereas the additional ambiguity parameters in AMM and CEU display greater variation and dispersion across task structures.

One possible interpretation is that greater structural complexity increases cognitive difficulty and decision noise, reducing the empirical benefit of additional model parameters. This interpretation is consistent with theories of bounded rationality and cognitive load. However, the experiment does not directly identify cognitive overload as the mechanism generating the observed model-fit pattern. Genuine heterogeneity in ambiguity attitudes and differences in the precision with which ambiguity parameters can be identified across tasks are also compatible with the results. The contribution of the paper is therefore not to establish a particular cognitive mechanism, but to show that the relative performance of models with explicit ambiguity parameters changes across differently structured ambiguous choice environments. This finding cautions against assuming that greater decision complexity necessarily implies that a more parameterized behavioural model will provide a better empirical description.

2. Related Literature

Ambiguity refers to situations in which the probabilities of possible outcomes are not known with certainty. The classic Ellsberg paradox demonstrates that individuals often distinguish between risky prospects with known probabilities and ambiguous prospects with uncertain probabilities, in ways that cannot be accommodated by standard Expected Utility (EU) theory (Ellsberg, 1961; Savage, 1954). This evidence motivated the development of models that explicitly incorporate ambiguity attitudes.

Several influential models provide alternative representations of behaviour under ambiguity. Choquet Expected Utility (CEU) relaxes the additivity of probabilities and evaluates uncertain events using capacities (Schmeidler, 1989). MaxMin Expected Utility represents ambiguity through a set of possible probability distributions and evaluates an act according to its worst-case expected utility (Gilboa & Schmeidler, 1989). The α MaxMin (AMM) model generalizes this approach by allowing decision-makers to place different weights on the best- and worst-case expected utilities (Ghirardato et al., 2004). Other approaches, such as the Smooth Model of Ambiguity Attitudes, provide further ways to separate beliefs about uncertainty from attitudes towards ambiguity (Klibanoff et al., 2005). In the empirical specifications considered in this paper, these models provide richer descriptions of ambiguity attitudes than EU but require additional ambiguity parameters to be identified from observed choices.

A separate literature examines how the structure and complexity of decision problems affect behaviour. Complexity can arise from several features of a choice environment, including the number of possible outcomes, the number of stages through which uncertainty is resolved,

and the cognitive demands associated with evaluating alternatives. More generally, bounded-rationality approaches suggest that individuals may simplify complex decision problems when fully processing all available information is costly (Gabaix, 2014). Experimental evidence also indicates that cognitive load can affect economic decision-making, including numerical performance and choices under risk (Deck & Jahedi, 2015). These findings suggest that increasing structural complexity may affect not only observed choices but also the precision with which richer behavioural models can be estimated.

The relationship between ambiguity and complexity has received comparatively less attention. Existing experimental studies have shown that ambiguity attitudes can depend on how ambiguous events and outcomes are structured. For example, Chew et al. (2017) show that behaviour under partial ambiguity depends on the position of an ambiguous event within the outcome distribution. Abdellaoui et al. (2011) develop methods for measuring source-dependent attitudes toward uncertainty across a rich domain of uncertain prospects, demonstrating substantial variation in attitudes across sources of uncertainty. These studies indicate that ambiguity attitudes need not be invariant to the structure or source of uncertainty. However, they do not directly address whether increasing the structural complexity of ambiguous choice problems changes the relative empirical performance of alternative models of ambiguity.

This paper contributes to these literatures by examining model performance across ambiguous choice environments that differ along two dimensions of structural complexity: the number of possible outcomes and the number of decision stages. The experiment compares EU with two models explicitly incorporating ambiguity attitudes, AMM and CEU. Rather than assuming that structurally richer ambiguous choices necessarily require more flexible behavioural models, we ask whether the relative empirical performance of these models changes as the choice environment becomes more complex. This approach complements previous work on ambiguity attitudes by focusing on the relationship between task structure and model performance, while remaining agnostic about the specific cognitive mechanism generating any observed differences.

3. Experimental Design

There were six sessions with 100 participants in total from the University of York. On average, each session lasted around 1 hour, and participants received £16.83 pp including £2.50 show-up fee. The total payment of the experiment was £1661.01¹. During the experiment, participants were asked to answer a series of allocation problems over four

¹ I gratefully acknowledge Prof. John Hey's generous individual funding for this experiment.

parts: Part RL, Part ALA, Part ALB and Part ALC. As this experiment mainly applied the allocation method, it is necessary to start with an explanation of how it is used to elicit preferences.

3.1 Allocation method

Participants are offered a certain number of tokens, F , as their endowment, and are asked to allocate it between two options with exchange rates between money and tokens. To demonstrate its application in preference elicitation, one of the allocation problems in Part RL will be explained in detail below. Options will be referred as colours.

For this problem, participants need to decide how to allocate $F = 100$ tokens between Red and Yellow. The probability of Red happening is p which is 70%, and that of Yellow happening is $1-p = 30\%$. The exchange rate of Red, E_1 , is £0.06 so one token allocated to Red is worth 6 pence, and one token allocated to Yellow is worth 8 pence (i.e. $E_2 = £0.08$). If this problem is chosen for determining participants' payments for this part, the computer will choose either Red or Yellow. Suppose a participant allocates X tokens to Red: she/he will get XE_1 if Red is played out by the computer. Otherwise, she/he will receive $£(F-X)E_2$. Figure 1 displays the software interface of this problem. In this case, $X = 50$, so the participant can get the amount of money allocated to Red, £3 = 50×0.06 , with chance 70% and get £4 which is the amount of money allocated to Yellow with chance 30%. Participants can choose their allocation by moving the slider. It should be noted that changing the allocation only affects the amounts of money allocated to Red and Yellow but not probabilities and exchange rates. The right-hand side histogram in Figure 1 visualizes the allocation. The horizontal axis shows the chances, and the vertical axis shows the amounts of money, so the chance of Red being chosen is fixed and shown by the horizontal distance of the Red rectangle – in this case 70%. Similarly, the chance of getting the money value of the tokens allocated to Yellow is 30%. Moving the slider will change heights of the rectangles change solely, but not their width. Probabilities of Red happening and Yellow happening, and exchange rates of them only change between problems. An allocation indicates a mixture of Red and Yellow. Participants can control how much risk they would expose to by changing the allocation. For example, an extreme risk-averse person may follow this half-allocation strategy to guarantee she/he can always get positive payments. If this problem were to be chosen by the computer at the end of Part RL, the lottery in Figure 1 would be played out. The spinner would be spun by the computer, and where it stops determines the participant's payoff for this problem (Figure 2).

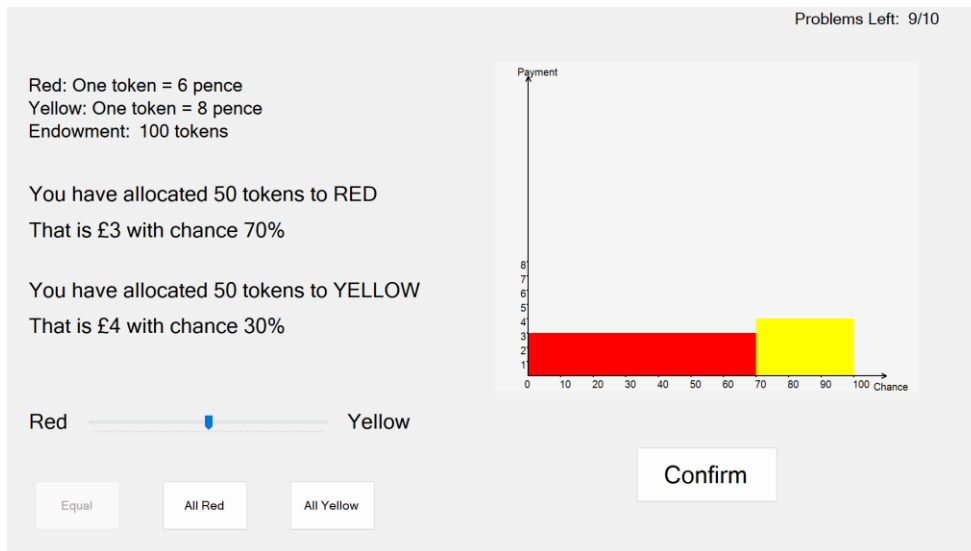


Figure 1. An example of a problem in the Risk part with an equal split allocation.

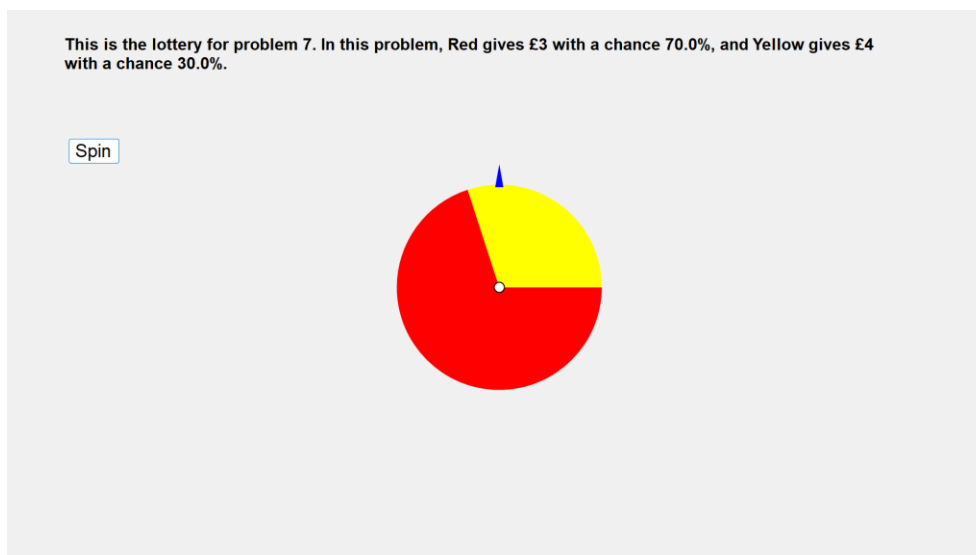


Figure 2. The computer choosing the problem

3.2 The Experiment

The experiment is in four parts: Part RL; Part ALA; Part ALB; Part ALC. In each part there are several allocation problems: 10 in Part RL; 10 in Part ALA; 15 in Part ALB; and 15 in Part ALC. After each part, the computer chooses a problem at random, and the payoff on that problem is the participant's payment for that part.

Overall, Part RL contains the simplest risk allocation problems, and the remaining three parts involve ambiguity which is reflected in the lack of knowledge of probabilities. **Participants are informed the lowest probability and the highest probability of a colour happening.** The

complexity of problems increases over parts from ALA to ALC in terms of two dimensions: the number of possible outcomes and the number of stages. These four parts are discussed below.

i. Part RL (Risky Lottery)

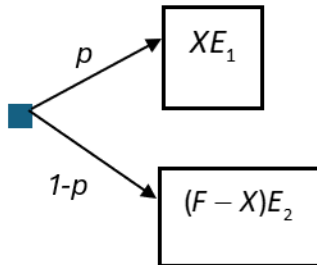


Figure 3. Game tree of questions in Part RL

To summarise, this is a one-stage risky lottery. A decision-maker (DM) is offered the endowment F and asked to decide how many tokens allocated to Red X with the exchange rate E_1 , and Yellow $(F-X)$ with the exchange rate E_2 . The probability of Red happening is p and that of Yellow happening is $1-p$. If Red happens, the DM will get XE_1 . If Yellow happens, the DM will get $(F-X)E_2$. Figure 1 has shown the software interface of this part.

ii. Part ALA (Ambiguous Lottery A)

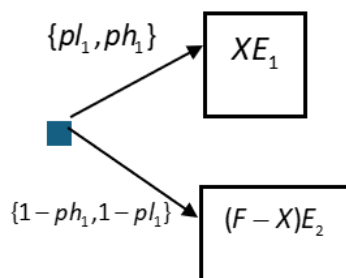


Figure 4. Game tree of questions in Part ALA

This is a one-stage ambiguous lottery. A DM is offered the endowment F and asked to decide how many tokens allocated to Red X with the exchange rate E_1 , and Yellow $(F-X)$ with the exchange rate E_2 . The DM is only told **two possible values** to represent the objective probability of Red happening: the highest probability, ph_1 , and the lowest probability, pl_1 . Therefore, the highest probability of Yellow happening is $1-pl_1$ and the lowest probability of

Yellow happening is $1-ph_1$. If Red happens, the DM will get XE_1 . If Yellow happens, the DM will get $(F-X)E_2$.

iii. Part ALB (Ambiguous Lottery B)

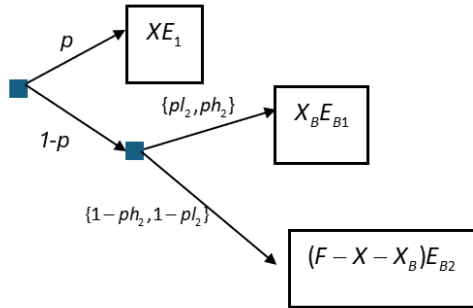


Figure 5. Game tree of questions in Part ALB

This is a **two-stage** ambiguous lottery. A DM is given an endowment of F tokens to allocate between Red, with probability of Red happening p , and another event which has the residual probability of $1-p$ of happening. Denote the allocation to Red by X . There is an exchange rate E_1 between tokens and money for Red. If Red happens, then the DM will get paid XE_1 . If Red does not happen, the DM will have to take another allocation decision: out of the remainder of the endowment $(F-X)$, the DM must choose X_B to allocate to Blue, which would occur **with either the highest probability, ph_2 , or the lowest probability, pl_2** - if it does occur the DM will get paid X_BE_{B1} ; where E_{B1} is the exchange rate between tokens and money for Blue; if it does not occur, the DM will get paid $(F-X-X_B)E_{B2}$ where E_{B2} is the exchange rate between tokens and money for Purple.

iv. Part ALC (Ambiguous Lottery C)

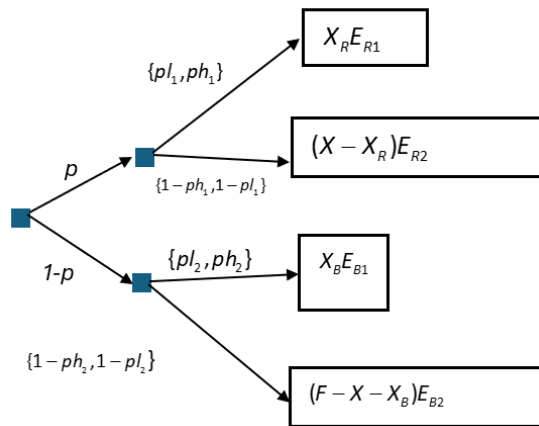


Figure 6. Game tree of questions in Part ALC

This is a **two-stage** lottery with **two** ambiguous branches. A DM is given an endowment of F tokens to allocate between the event (Red or Orange) with probability p of happening, and another event (Blue or Purple) which has the residual probability of $1-p$ of happening. Denote the allocation to Red or Orange by X . If Red or Orange does happen, the DM is asked to take another allocation decision: based on the endowment X to choose X_R to allocate to Red which would occur **with either the highest probability, ph_1 , or the lowest probability, pl_1** , with the exchange rate E_{R1} . If Red happens, the DM will get $X_R E_{R1}$. If Red does not happen, the DM will get paid $(X - X_R) E_{R2}$, where E_{R2} is the exchange rate between tokens and money for Orange. Similarly, If Red or Orange does not happen that is, (Blue or Purple does happen), the DM is asked to take another allocation decision: out of the remainder of the endowment $(F - X)$, the DM must choose X_B to allocate to Blue which would occur **with either highest probability, ph_2 , or the lowest probability, pl_2** , – if it does occur the DM gets paid $X_B E_{B1}$ where E_{B1} is the exchange rate between tokens and money for Blue; if it does not occur, the DM gets paid $(F - X - X_B) E_{B2}$ where E_{B2} is the exchange rate between tokens and money for Purple.

The software interfaces of these three ambiguous lottery parts are displayed in Figure 7, 8 and 9 respectively.

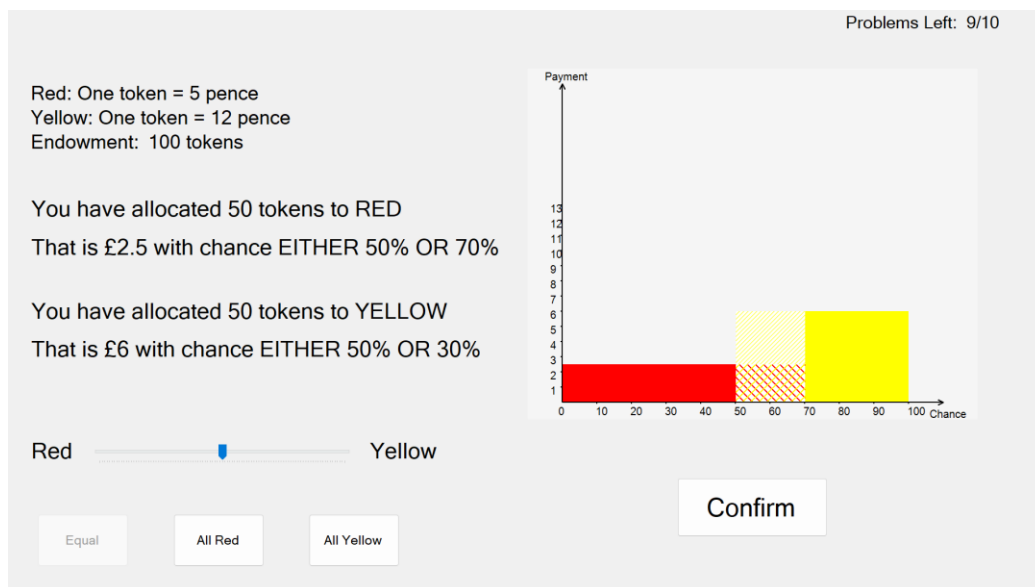


Figure 7. Software interface of Part ALA

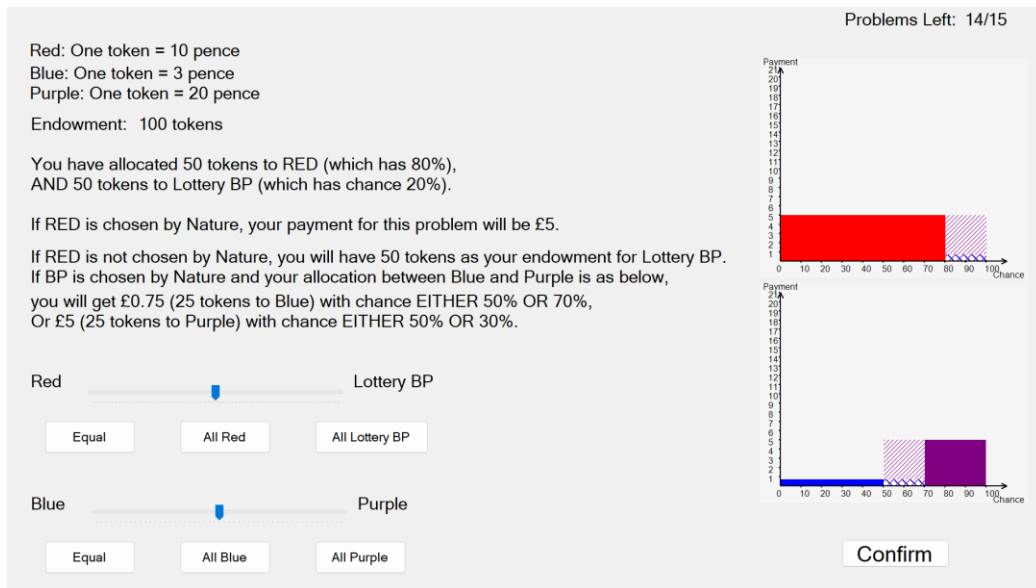


Figure 8. Software interface of Part ALB

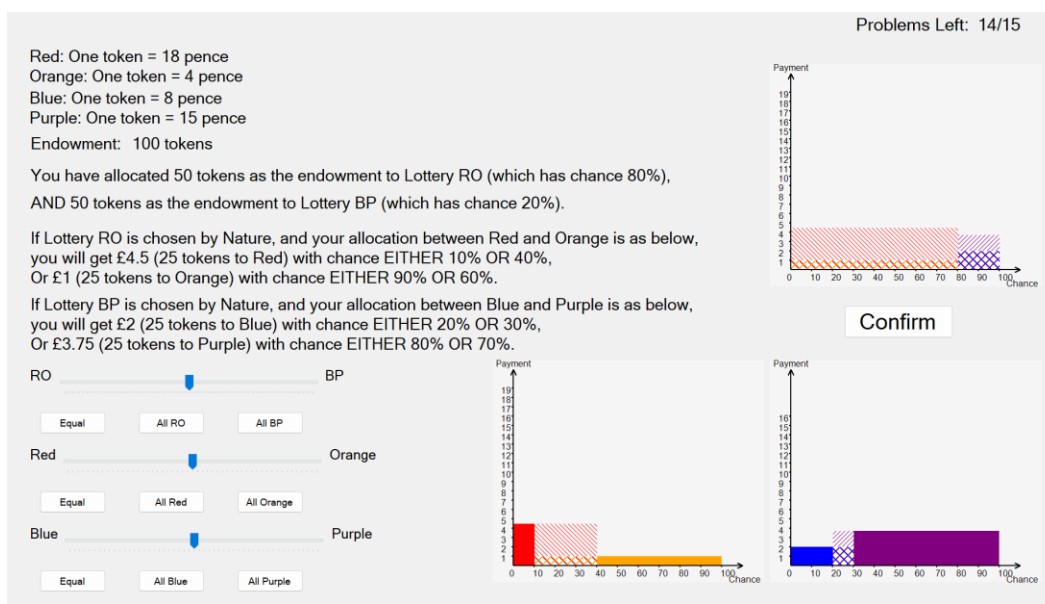


Figure 9. Software interface of Part ALC

4. Theoretical Framework

The experiment is designed to examine decision-making under risk and ambiguity across choice environments with different levels of structural complexity. To compare the empirical performance of the alternative models, we derive the optimal allocation predicted by each model for each experimental problem and compare these predictions with participants' observed allocations. The model-implied optimal allocations depend on the relevant preference parameters and on the probabilities and exchange rates specified in each

problem. We are going to assume that these latter are: (Subjective) Expected Utility ((S)EU), α MaxMin (AMM) model and Choquet Expected Utility (CEU). The theoretical optimal allocation X for each problem in the experiment can be calculated based on these models.

To allow for stochastic variation in observed decisions, we adopt an additive error specification in the spirit of the Fechnerian stochastic choice approach commonly used in experimental decision theory (Hey & Orme, 1994). The actual allocation (AA) is assumed to equal the optimal allocation (X) plus noise with a normal distribution $(0, \sigma^2)$ in each problem. Such noise would change over parts and be different among individuals: $AA_{ijk} = X_{ijk} + \varepsilon_{ik}$, where $\varepsilon_{ik} \sim (0, \sigma_{ik}^2)$, i indicates the subject number $i = \{1, 2, \dots, 100\}$, j represents the problem number within each part and k here indicates the part $k = \{RL, ALA, ALB, ALC\}$. The optimal allocation is a function of key parameters from these models with given experiment parameters - probabilities and exchange rates – as we discuss below.

The methodology is as follows:

Estimation is conducted separately for each participant, experimental part, and model using maximum likelihood. To address identification issues, we adopt a sequential estimation procedure. We first use the observations from Part RL to estimate the CRRA risk-aversion parameter r . For the ambiguity models, the estimated r from Part RL is then held fixed when estimating the ambiguity parameters. Specifically, we estimate α for the AMM model and γ for the CEU model separately using the observations from Parts ALA, ALB, and ALC. The corresponding noise parameter is estimated separately for each participant, part, and model.

An inevitable by-product of the maximum likelihood estimation is, for each subject, each part and each model, the value of the maximised log-likelihood. This is an obvious measure of the goodness-of-fit of that model for that subject for that part and that model. To compare these goodness-of-fit across models and parts we need to work with a corrected Akaike Criterion considering differences in the number of parameters and the number of observations.

4.1 Model

The underlying utility function is assumed to be the CRRA model: $u(x, r) = \frac{x^{1-r}}{1-r}$ for $r \geq 0$ and $r \neq 1$; $u(x) = \ln(x)$ for $r = 1$. For the former, as a more general case, r is the CRRA index of risk

aversion and u takes the form $u(x,r) = \frac{x^{1-r}}{1-r}$ and $u'(x,r) = x^{-r}$. For the latter, $u(x) = \ln(x)$ and $u'(x) = x^{-1}$. To find the optimal allocation, we took the first-order derivative of the expected utility with respect to X and put the answer equal to 0.

a. (S)EU model

It is the standard expected utility theory with the key parameter r from the CRRA model. If $r = 0$, individuals would be risk neutral. If $r > 0$, they would be risk averse, and the larger the r is, the more risk averse the DM. Otherwise, they are risk seeking. In an ambiguous situation, it becomes the subjective EU model which assumes that individuals make decisions based on their *subjective* beliefs about the probabilities (Savage, 1954). As the target is to estimate r and check if r changes over three parts (Part ALA, Part ALB and Part ALC), the subjective probability in the ambiguous case was simply assumed to be the average of the highest probability, ph , and the lowest probability, pl .

b. AMM model

This model focuses on individual decision making under ambiguity. Developing from the MaxMin model (Gilboa & Schmeidler, 1989), it introduces a new parameter α to generalize this story (Ghirardato et al., 2004). The AMM model assumes that people consider both the best case and worst case in a weighted way. $\alpha \in [0,1]$ is not only the weighted average parameter but also the ambiguity aversion index. If $\alpha = 1$, people would be fully pessimistic, only looking at worst cases, and the AMM model reduces to the MaxMin model. In contrast, if $\alpha = 0$, only best cases matter, and the model reduces to the MaxMax model. When $\alpha = 0.5$, individuals weight these two extreme situations equally. Combining the setting of the experiment and the assumption in the EU model, under such situation, the AMM model reduces to the EU model mathematically. Individuals are treated as ambiguity neutral, as they behave similarly to a standard EU maximiser with a 'representative' prior $p = \frac{pl + ph}{2}$. Generally, if $0 \leq \alpha < 0.5$, people would be ambiguity seeking; if $0.5 < \alpha \leq 1$, they would be ambiguity averse.

c. CEU model

The standard EU model relies on additive probabilities: for disjoint events A and B , $P(A \cup B) = P(A) + P(B)$. CEU relaxes this additivity requirement by evaluating events using a capacity $v(\cdot)$ rather than an additive probability measure. It states that under ambiguity, additivity is not necessary, and individuals are assumed to rely on capacities $v(\cdot)$ to evaluate

events instead of probabilities (Schmeidler, 1989). We applied the power capacity function to capture how people distort their subjective probabilities, which are same as those in the EU model, in ambiguous situations: $v(p) = p^\gamma$, where $\gamma > 0$. In addition to be treated as a probability distortion parameter, γ reflects the degree of ambiguity aversion. If $\gamma > 1$, individuals would be ambiguity averse. If $1 > \gamma > 0$, they would be ambiguity lovers. Otherwise, this model collapses to the (S)EU model, and individuals would be seen as ambiguity neutral. This function is commonly used. Given a two-positive-outcome lottery, the power capacity will be identical to the probability weighting function applied in Cumulative Prospect Theory (Tversky & Kahneman, 1992). Also, if probabilities become objective, the CEU model with the power capacity can be directly treated as Rank-Dependent Utility Theory.

4.2 Hypothesis

Compared with the AMM and CEU models, (S)EU provides a relatively parsimonious representation of behaviour because it does not introduce an additional parameter to capture ambiguity attitudes. A natural intuition is that more complex choice problems may require more flexible behavioural models to describe individual behaviour. However, an alternative possibility is that the empirical benefit of additional model flexibility decreases as ambiguous choice problems become more complex. We therefore examine whether the relative goodness-of-fit of (S)EU, AMM, and CEU changes with the structural complexity of the ambiguous lotteries.

One motivation for this possibility comes from bounded rationality. Gabaix (2014) argues that when decision environments are complex, individuals may simplify the way in which they process available information. Experimental evidence also shows that cognitive load can affect economic decision-making and increase variability in observed behaviour (Deck & Jahedi, 2015). Applied to the present setting, greater structural complexity may therefore make behaviour more difficult to capture through additional ambiguity parameters. This provides one possible reason why a relatively parsimonious model such as (S)EU could perform as well as, or better than, more flexible ambiguity models in complex choice environments.

A related statistical consideration is parameter identification. AMM and CEU introduce additional ambiguity parameters, and estimating these parameters precisely requires sufficient variation and information in observed choices. If choices become more variable as task complexity increases, or if the available choice data provide limited information for

identifying the additional parameters, greater model flexibility may not translate into better penalized model fit. Cognitive difficulty is therefore one possible explanation for the hypothesized pattern, but it is not the only one; differences in ambiguity attitudes or parameter identification across task structures could generate similar patterns.

Accordingly, our hypothesis is that increasing structural complexity does not necessarily improve the relative fit of more parameterized ambiguity models; relatively parsimonious models such as (S)EU may provide an equally good or better empirical description of behaviour as ambiguous choice problems become more complex.

5. Econometric Method

A key objective of the econometric analysis is to investigate whether estimated risk and ambiguity attitudes and the relative fit of alternative models differ across experimental parts as the structural complexity of the ambiguous lotteries increases. We estimate the relevant model parameters separately for each participant and experimental part using maximum likelihood estimation. Table 1 summarizes the parameters estimated under each model.

As an illustration, consider the simplest risk-averse case to show how model predictions and observed allocations can be used to infer the value of the parameter r . In Part RL, the expected utility is $EU(X) = pu(XE_1, r) + (1 - p)u((F - X)E_2, r)$. After differentiating it w.r.t. X and

setting the first-order condition equal to zero, $X = \frac{A}{A + B}F$, where $A = (1 - p)^{-1/r} E_2^{1-1/r}$ and

$B = p^{-1/r} E_1^{1-1/r}$. As values of p , E_1 and E_2 are known, the optimal allocation X can be expressed as a function of r , denoted $X(r)$. Based on each participant's observed allocations and the corresponding model-implied optimal allocations, maximum likelihood estimation is used to obtain the best-fitting value of r . The Technical Appendix provides further details on the derivation of optimal allocations under the different models and experimental parts, including the risk-neutral and risk-seeking cases.

Model	EU	RN ²	AMM	CEU
Number of parameters	8	4	8 ³	8
Part RL	(r_{RL}, σ_{rRL})	σ_{rRL}	(r_{rRL}, σ_{rRL})	(r_{RL}, σ_{rRL})
Part ALA	(r_{ALA}, σ_{rALA})	σ_{rALA}	$(\alpha_{ALA}, \sigma_{\alpha ALA})$	$(\gamma_{ALA}, \sigma_{\gamma ALA})$
Part ALB	(r_{ALB}, σ_{rALB})	σ_{rALB}	$(\alpha_{ALB}, \sigma_{\alpha ALB})$	$(\gamma_{ALB}, \sigma_{\gamma ALB})$
Part ALC	(r_{ALC}, σ_{rALC})	σ_{rALC}	$(\alpha_{ALC}, \sigma_{\alpha ALC})$	$(\gamma_{ALC}, \sigma_{\gamma ALC})$

Table 1. Parameters estimated by model and experimental part

6. Results

The specific estimation results of parameters under different models are presented on the [EXEC website](#). Because the number of observations available for each participant is relatively small and differs across experimental parts, we use the small-sample corrected Akaike Information Criterion (AICc) for model comparison. AICc is computed based on the regular AIC which is calculated by $2k - 2\text{Loglikelihood}$, where k is the number of parameters of the model and Loglikelihood is the maximised log-likelihood. With AIC, AICc is given by

$$AICc = AIC + \frac{2k(k+1)}{n-k-1}, \text{ where } n \text{ indicates the sample size. The lower the value of AICc, the}$$

corresponding model fits better. Figure 10 shows the fitness ranking results over parts⁴. For example, the number ‘52’ in Part ALA indicates that, among the 100 participants, EU provides the best fit according to AICc for 52 participants. Overall, EU is the most frequently selected model in all three parts, providing the best fit for 52 participants in ALA, 60 in ALB, and 54 in ALC. CEU also performs relatively well in ALC, providing the best fit for 42 participants. In contrast, AMM is selected as the best-fitting model for only 4 participants in ALC. Chi-square goodness-of-fit tests show that the distribution of best-fitting models differs significantly from an equal-frequency benchmark in all three parts (ALA, $\chi^2(2) = 15.92$,

² RN indicates the risk neutral case based on the EU model ($r = 0$). It is considered as the nested benchmark of EU.

³ For the EU model, r and the corresponding noise parameter σ_r are estimated separately in each experimental part. For the AMM model, r is first estimated using Part RL and then held fixed when estimating α and the corresponding noise parameter σ_α separately in Parts ALA, ALB, and ALC. The same sequential procedure is applied to the CEU model, with γ and the corresponding noise parameter σ_γ estimated separately in each ambiguous-lottery part conditional on the estimated r from Part RL. This sequential estimation procedure is adopted to avoid identification problems that arise when r , the ambiguity parameter α or γ , and the noise parameter are estimated simultaneously.

⁴ As the complexity only changed in ambiguous lotteries, the comparison of model fitness focuses on Part ALA, ALB and ALC. The structure of ambiguous lotteries is more complex from ALA to ALC.

$p < 0.001$; ALB, $\chi^2(2) = 32.00, p < 0.001$; and ALC, $\chi^2(2) = 40.88, p < 0.001$).

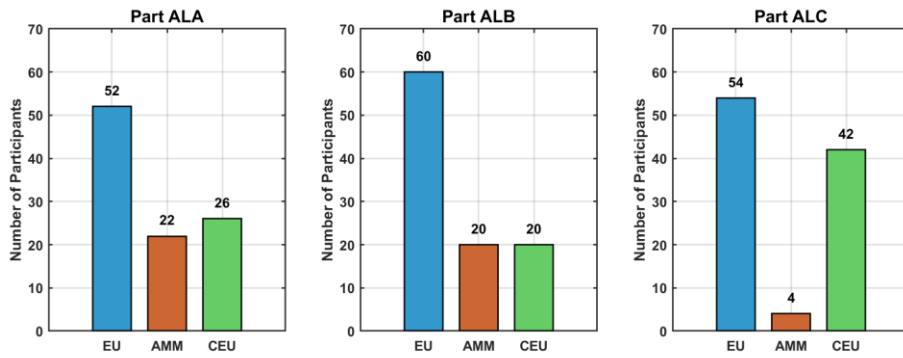


Figure 10. Distribution of the best fitting model over parts

To further assess whether EU provides a better fit than the alternative models, we compare participant-level AICc values. For each comparison, we calculate the $\Delta AICc = AICc_{EU} - AICc_{comparison}$, such that a negative difference indicates a better fit of EU. We first conduct one-tailed paired t-tests against the alternative that the mean difference is negative. Because participant-level AICc differences may not be normally distributed and may be affected by large differences for a subset of participants, we additionally conduct one-tailed Wilcoxon signed-rank tests as a robustness check. Table 2 reports the mean and median AICc differences, the number of participants for whom EU yields a lower AICc, and the results of both tests.

The results provide the strongest evidence in favour of EU in Part ALB. EU yields a lower AICc than AMM for 71 participants and than CEU for 74 participants, with both the paired t-tests and Wilcoxon signed-rank tests rejecting the null hypotheses ($p < 0.001$). In Part ALC, EU also performs significantly better than AMM (Wilcoxon $p < 0.001$), with a lower AICc for 78 participants. In contrast, there is no significant difference between EU and CEU in ALC (Wilcoxon $p = 0.291$), and EU has a lower AICc for 57 participants.

The evidence for Part ALA is more sensitive to the statistical test employed. The paired t-tests do not reject the null for either EU versus AMM ($p = 0.080$) or EU versus CEU ($p = 0.116$). However, the corresponding Wilcoxon signed-rank tests favour EU over both AMM ($p < 0.001$) and CEU ($p = 0.001$). EU also yields a lower AICc for 65 and 69 participants, respectively. Thus, rather than concluding that EU does not outperform the ambiguity models in ALA, the results indicate that the evidence depends on the distributional assumptions underlying the test.

As discussed above, RN is treated as the nested risk-neutral benchmark of EU rather than as a separate competing behavioural model. We therefore report EU–RN comparisons separately as a supplementary benchmark. EU yields a lower AICc than RN for 95, 97, and 93 participants in ALA, ALB, and ALC, respectively, and both the paired *t*-tests and Wilcoxon signed-rank tests favour EU in all three parts (all $p < 0.001$).

Part	Comparison	Mean $\Delta AICc$	Median $\Delta AICc$	EU Lower AICc	t-test p	Wilcoxon p
ALA	EU-AMM	-5.356	-0.521	65	0.080	<0.001
ALA	EU-CEU	-4.559	-0.483	69	0.116	0.001
ALA	EU-RN	-203.987	-250.144	95	<0.001	<0.001
ALB	EU-AMM	-13.305	-5.298	71	<0.001	<0.001
ALB	EU-CEU	-12.132	-6.955	74	<0.001	<0.001
ALB	EU-RN	-117.312	-59.728	97	<0.001	<0.001
ALC	EU-AMM	-40.904	-13.963	78	0.002	<0.001
ALC	EU-CEU	-8.853	-1.484	57	0.186	0.291
ALC	EU-RN	-120.807	-82.411	93	<0.001	<0.001

Table 2. Participant-level comparisons of AICc across models

*Notes: $\Delta AICc = AICc_{EU} - AICc_{comparison}$; negative values favour EU. “EU lower AICc” reports the number of participants, out of 100, for whom EU yields a lower AICc than the comparison model. *p*-values are one-sided, corresponding to the alternative that EU has lower AICc. RN is reported as a nested risk-neutral benchmark rather than as a separate substantive competing model.*

Because AICc explicitly penalizes model complexity, the poorer AICc performance of a more parameterized model should not by itself be interpreted as evidence of overfitting. The results instead indicate differences in relative penalized fit across the competing models. In particular, the relative benefit of allowing additional ambiguity parameters varies across the three ambiguous-lottery parts. Although cognitive difficulty and decision noise provide one possible explanation for this pattern, the present design does not independently identify the mechanism underlying the differences in model performance. Alternative explanations, including heterogeneity in ambiguity attitudes and differences in parameter identification across task structures, are also consistent with the observed results.

The observed parameter distributions provide additional descriptive evidence on how model estimates vary across parts. Figures 11-13 report the relevant parameter estimates for all 100 participants, irrespective of which model provides the lowest AICc for a given participant. Thus, the distributions are intended to describe the estimated parameters under each model for the full sample rather than only for participants best fitted by that model. Distributions of the estimated risk-aversion parameter *r* under EU are broadly similar across the three parts involving ambiguous lotteries (Figure 11). By contrast, the distributions of the

ambiguity parameters α and γ differ more noticeably across parts (Figures 12 and 13). In particular, their distributions in Part ALA are more concentrated around intermediate values, whereas Parts ALB and ALC display greater dispersion and more estimates close to parameter boundaries.

The relative stability of r across parts suggests that estimated risk attitudes are less sensitive to changes in task structure than the additional ambiguity parameters. However, the more dispersed and boundary-concentrated estimates of α and γ in the more complex parts should not be interpreted uniquely as evidence of noise. Such estimates may reflect genuinely strong ambiguity attitudes, imprecise parameter identification, or both⁵, although the present data do not allow these explanations to be distinguished conclusively.

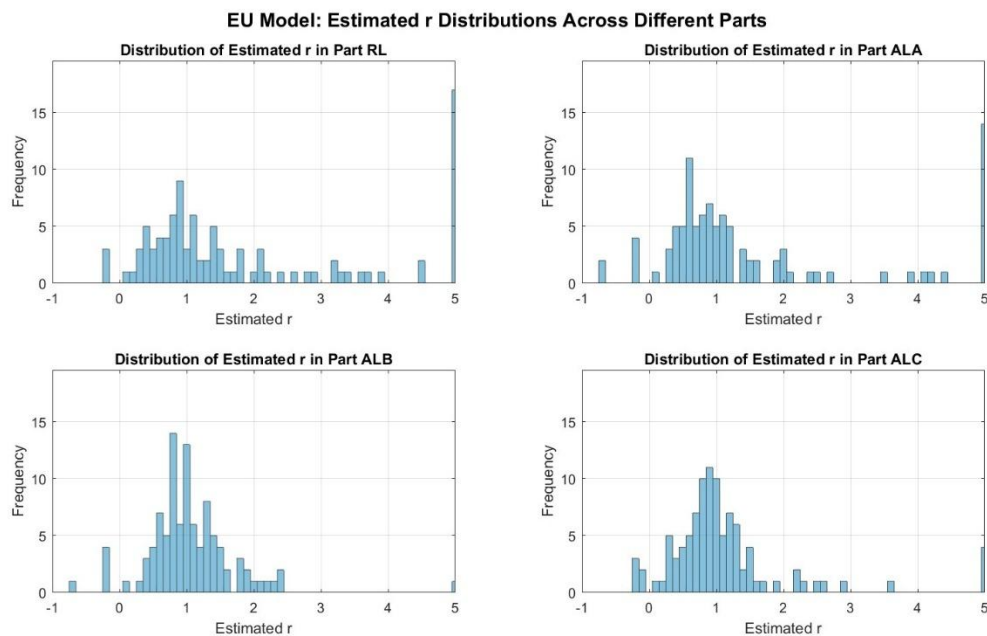


Figure 11. Distribution of estimated r across parts for all participants ($N = 100$)

⁵ For example, some estimates of α are concentrated near 0 or 1, which under the AMM specification corresponds to very strong pessimistic or optimistic weighting of the lower and higher probability scenarios. Similarly, some estimates of γ lie close to the boundary of its parameter space. Such estimates are consistent with strong ambiguity attitudes, but may also reflect imprecise parameter identification; the present analysis cannot distinguish these interpretations.

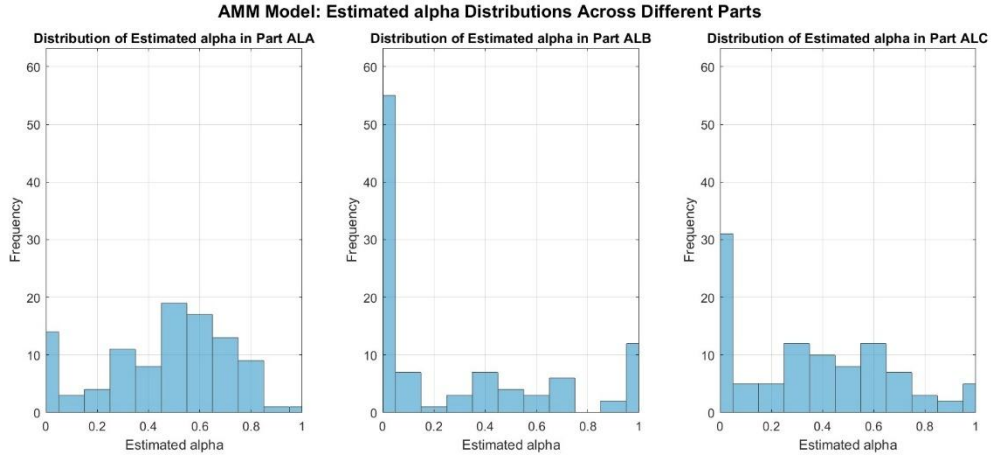


Figure 12. Distribution of estimated α across parts for all participants ($N = 100$)

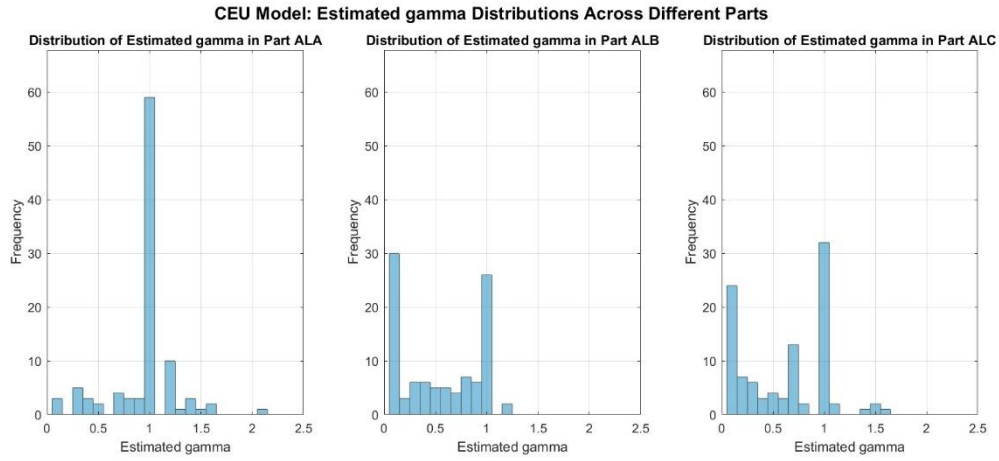


Figure 13. Distribution of estimated γ across parts for all participants ($N = 100$)

We further examine the estimated ambiguity parameters using a series of t-tests reported in Tables 3 and 4. To provide information on the precision of the estimated mean parameters, we also report 95% confidence intervals. Because these tables contain 12 hypothesis tests in total, we apply the Benjamini–Hochberg false discovery rate (BH-FDR) correction jointly across all 12 tests and report both the unadjusted and BH-FDR-adjusted p-values. The correction does not change any of the original significance decisions.

In Part ALA, the null hypotheses $\alpha = 0.5$ and $\gamma = 1$ are not rejected (p-value = 0.056, BH-FDR p-value = 0.061, and p-value = 0.147, BH-FDR p-value = 0.147, respectively). The corresponding 95% confidence intervals for the mean parameter estimates are [0.401, 0.501] for α and [0.890, 1.017] for γ , both of which contain their respective ambiguity-neutral benchmark values. Thus, the estimates in Part ALA are statistically consistent with the ambiguity-neutral benchmarks, but this should not be interpreted as establishing

ambiguity neutrality. In particular, the benchmark $\alpha = 0.5$ lies close to the upper bound of its confidence interval, indicating that the evidence regarding neutrality under AMM is relatively weak. In Parts ALB and ALC, by contrast, the confidence intervals exclude the corresponding ambiguity-neutral benchmarks, and both $\alpha = 0.5$ and $\gamma = 1$ are rejected after the BH-FDR correction (all adjusted $p < 0.001$).

Test	Part	Mean α	Std α	95% CI for mean	t-statistic	p-value	BH-FDR p
$H_0: \alpha = 0; H_1: \alpha > 0$	ALA	0.451	0.254	[0.401, 0.501]	17.782	<0.001	<0.001
$H_0: \alpha = 0.5; H_1: \alpha \neq 0.5$					-1.936	0.056	0.061
$H_0: \alpha = 1; H_1: \alpha < 1$					-21.654	<0.001	<0.001
$H_0: \alpha = 0; H_1: \alpha > 0$	ALB	0.262	0.360	[0.191, 0.334]	7.295	<0.001	<0.001
$H_0: \alpha = 0.5; H_1: \alpha \neq 0.5$					-6.602	<0.001	<0.001
$H_0: \alpha = 1; H_1: \alpha < 1$					-20.498	<0.001	<0.001
$H_0: \alpha = 0; H_1: \alpha > 0$	ALC	0.344	0.308	[0.283, 0.405]	11.154	<0.001	<0.001
$H_0: \alpha = 0.5; H_1: \alpha \neq 0.5$					-5.055	<0.001	<0.001
$H_0: \alpha = 1; H_1: \alpha < 1$					-21.263	<0.001	<0.001

Table 3. Hypothesis tests for the estimated AMM parameter α

Test	Part	Mean γ	Std γ	95% CI for mean	t-statistic	p-value	BH-FDR p
$H_0: \gamma = 1; H_1: \gamma \neq 1$	ALA	0.953	0.320	[0.890, 1.017]	-1.461	0.147	0.147
$H_0: \gamma = 1; H_1: \gamma \neq 1$	ALB	0.551	0.379	[0.476, 0.626]	-11.861	<0.001	<0.001
$H_0: \gamma = 1; H_1: \gamma \neq 1$	ALC	0.618	0.410	[0.537, 0.699]	-9.317	<0.001	<0.001

Table 4. Hypothesis tests for the estimated CEU parameter γ

Notes: The 95% confidence intervals refer to the mean parameter estimates across participants. Raw p-values are reported together with Benjamini–Hochberg false discovery rate adjusted p-values. The BH-FDR adjustment is applied jointly across the 12 hypothesis tests reported in Tables 3 and 4.

The preceding model comparisons assess the relative fit of EU against the alternative models. We next examine the absolute goodness-of-fit of the EU model using McFadden's pseudo- R^2 , defined as $R^2_{McFadden} = 1 - \frac{\ln L_{EU}}{\ln L_{RN}}$, where $\ln L_{EU}$ is the maximized log-likelihood under the EU model and $\ln L_{RN}$ is the log-likelihood under the risk-neutral specification ($r = 0$), which serves as the null model. Higher values indicate a greater improvement in fit relative to the risk-neutral benchmark.

Table 5 reports both aggregate and individual-level pseudo- R^2 results. The aggregate pseudo- R^2 , calculated after summing the log-likelihoods across participants, is 0.719 in Part ALA, 0.349 in Part ALB, and 0.275 in Part ALC. Because aggregate measures may conceal

heterogeneity in model fit across participants, we additionally calculate pseudo- R^2 separately for each of the 100 participants. The median individual-level pseudo- R^2 is 0.725 in ALA, 0.213 in ALB, and 0.200 in ALC, with corresponding interquartile ranges of [0.259, 0.825], [0.131, 0.298], and [0.123, 0.281], respectively. Figure 14 presents the full distributions.

The individual-level results show that the aggregate goodness-of-fit is not solely driven by a small subset of participants. This is particularly clear in Part ALA, where the first quartile of individual pseudo- R^2 is 0.259. In Parts ALB and ALC, the distributions are more concentrated at lower values and exhibit greater heterogeneity, including a small number of negative pseudo- R^2 values, indicating that for some participants the EU model does not improve upon the risk-neutral benchmark. Overall, the individual-level analysis therefore supports meaningful fit of the EU model for a substantial proportion of participants, while also showing that the strength of fit varies across participants and experimental parts.

Part	N	Aggregate pseudo- R^2	Mean	Median	Q1	Q3	Min	Max
ALA	100	0.719	0.551	0.725	0.259	0.825	0.000	0.898
ALB	100	0.349	0.261	0.213	0.131	0.298	-0.107	0.827
ALC	100	0.275	0.211	0.200	0.123	0.281	-0.459	0.842

Table 5. Aggregate and individual-level McFadden pseudo- R^2 for the EU model

Notes: Aggregate pseudo- R^2 is calculated from log-likelihoods summed across participants, whereas the remaining statistics summarize pseudo- R^2 calculated separately for each participant. Q1 and Q3 denote the 25th and 75th percentiles, respectively.

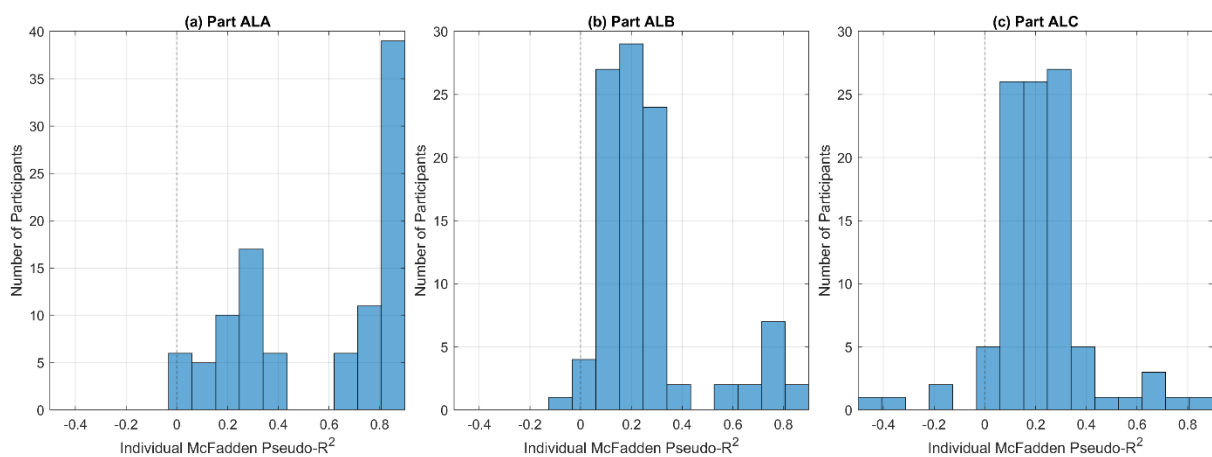


Figure 14. Distribution of individual-level Pseudo- R^2 for the EU model across parts

7. Conclusion

This study examines how the relative performance of alternative decision models changes as ambiguous choice problems become structurally more complex. Using an allocation experiment, complexity is varied along two dimensions: the number of possible outcomes and the number of decision stages, and behaviour is evaluated using EU, AMM, and CEU.

The model-comparison results provide qualified support for the idea that greater structural complexity does not necessarily favour more parameterized ambiguity models. EU is the most frequently selected model according to AICc in all three ambiguous-lottery parts. The strength of this result, however, varies across parts. In Part ALB, EU performs significantly better than both AMM and CEU according to both the paired t-tests and Wilcoxon signed-rank tests. In Part ALC, EU performs better than AMM, whereas its performance is not significantly different from CEU. In Part ALA, the conclusion depends partly on the statistical test employed. Thus, the evidence does not imply that EU universally dominates ambiguity models, but rather that allowing additional ambiguity parameters does not uniformly improve penalized model fit as the structure of the decision problem becomes more complex.

The parameter estimates provide a complementary perspective. The estimated risk-aversion parameter r under EU is relatively stable across the ambiguous-lottery parts, while the AMM parameter α and CEU parameter γ show greater variation and dispersion across tasks. In Part ALA, the mean ambiguity parameters are statistically consistent with their ambiguity-neutral benchmarks, although this should not be interpreted as establishing ambiguity neutrality. In Parts ALB and ALC, the ambiguity-neutral benchmarks are rejected. Moreover, the greater incidence of boundary estimates in the more complex parts may reflect genuinely strong ambiguity attitudes, imprecise parameter identification, or both. The present data do not allow these interpretations to be separated conclusively.

The individual-level goodness-of-fit analysis also reveals substantial heterogeneity. Although EU provides meaningful fit relative to the risk-neutral benchmark for a substantial proportion of participants, particularly in Part ALA, the strength of fit varies considerably across participants and parts. Some participants in ALB and ALC have negative individual pseudo- R^2 values, indicating that EU does not improve upon the risk-neutral benchmark for every individual. This reinforces the importance of complementing aggregate model comparisons with participant-level evidence.

Several limitations qualify the interpretation of these findings. First, the experimental design varies task structure across parts but does not independently manipulate or directly measure cognitive load. Consequently, cognitive difficulty, decision noise, or participant adaptation cannot be identified as the causal mechanisms behind the observed changes in model performance. Second, changes in estimated ambiguity parameters across tasks may reflect genuine preference heterogeneity, differences in parameter identification, or both. Third, the parts differ not only in complexity but also in their specific decision structures, so the effects of the number of outcomes and the number of stages cannot be fully isolated from other task characteristics. Finally, the analysis compares a particular set of structural models and functional forms; other models of ambiguity or alternative stochastic specifications may produce different rankings.

These limitations suggest several directions for future research. Experiments that independently manipulate cognitive load while holding the underlying ambiguous lottery fixed could provide a cleaner test of whether cognitive difficulty contributes to changes in relative model performance. Designs with repeated or counterbalanced task orders could also distinguish structural complexity from learning or adaptation. Increasing the number and variation of choices within each task would help assess whether boundary estimates reflect genuine preferences or weak parameter identification. Finally, comparing a broader set of ambiguity models and alternative stochastic choice specifications would establish whether the present model-ranking pattern generalizes beyond the specifications considered here.

Taken together, the results suggest a more qualified interpretation of “less is more.” The evidence does not establish that decision-makers necessarily adopt simpler cognitive strategies when faced with greater ambiguity and complexity. Rather, it shows that additional structural flexibility in a behavioural model does not automatically translate into better penalized empirical fit. Which model provides the most useful description of behaviour appears to depend on the structure of the choice environment, highlighting the value of evaluating model complexity empirically rather than equating greater behavioural sophistication with greater predictive success.

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